CS4451 & CS9548: Foundations of Machine Learning – Assignment #1

Due: 23:59, October 22, 2026

Instructions

Your submission should include: 1) a .pdf file that containing all the answers to the questions. For the questions requiring to plot figures, you should also include all the figures in the .pdf fie. 2) source code (e.g., .py files, python notebook, .m files...)

1 Refreshing Mathematics [25 points]

Let $w \in \mathbb{R}^n$ is an n-dimensional column vector, and $f(w) \in \mathbb{R}$ is a function of w. In Lecture 2, we have defined the *gradient* $\nabla f(w) \in \mathbb{R}^n$ and *Hessian matrix* $H \in \mathbb{R}^{n \times n}$ of f with respect to w.

(a) [5 points] Let $a, b \in \mathbb{R}^n$ be two n-dimensional vectors, and $a \circ b \in \mathbb{R}^n$ be the *element-wise* product of two vectors.[1] Let $f(w) = a^\top (w \circ b)$. Compute $\nabla f(w)$ using the definition of gradient.

(b) [5 points] Let $f(w) = \text{tr}(Bww^\top A)$, where $A, B \in \mathbb{R}^{n \times n}$ are squared matrices of size $n \times n$, and $\text{tr}(A)$ is the *trace* of the squared matrix A. Using the definition of gradient, compute $\nabla f(w)$.

(c) [5 points] Let $f(w) = \text{tr}(Bww^\top A)$. Compute the *Hessian matrix* H of f with respect to w using the definition.

(d) [5 points] If $A = \begin{bmatrix} 1 & -2 \\ -2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix}$. Is $f(w)$ a convex function? (Hint: you may use Python/Matlab/R for this question.)

(e) [5 points] In Lecture 5, we have define the sigmoid function: $\sigma(a) = \frac{1}{1+e^{-a}}$. Let $f(w) = \log(\sigma(w^\top x))$, where log is the natural logarithmic function. Compute $\nabla f(w)$ using the definition of gradient.

2 Linear and Polynomial Regression [50 points]

For this exercise, you will implement linear and polynomial regression in any programming language of your choice (e.g., Python/Matlab/R). The training data set consists of the features `hw1xtr.dat` and their desired outputs `hw1ytr.dat`. The test data set consists of the features `hw1xte.dat` and their desired outputs `hw1yte.dat`.

(a) [5 points] Load the training data `hw1xtr.dat` and `hw1ytr.dat` into the memory and plot it on one graph. Load the test data `hw1xte.dat` and `hw1yte.dat` into the memory and plot it on another graph.

(b) [10 points] Add a column vector of 1's to the features, then use the linear regression formula discussed in Lecture 3 to obtain a 2-dimensional weight vector. Plot both the linear regression line and the training data on the same graph. Also report the average error on the training set using Eq. (1).

$$err = \frac{1}{m} \sum_{i=1}^{m} (w^\top x_i - y_i)^2 \tag{1}$$

[1] See `https://en.wikipedia.org/wiki/Hadamard_product_(matrices)` for the definition and some examples. Note that a vector is a special case of a matrix.

(c) [5 points] Plot both the regression line and the test data on the same graph. Also report the average error on the test set using Eq. (1).

(d) [10 points] Implement the 2nd-order polynomial regression by adding new features x^2 to the inputs. Repeat (b) and (c). Compare the training error and test error. Is it a better fit than linear regression?

(e) [10 points] Implement the 3rd-order polynomial regression by adding new features x^2, x^3 to the inputs. Repeat (b) and (c). Compare the training error and test error. Is it a better fit than linear regression and 2nd-order polynomial regression?

(f) [10 points] Implement the 4th-order polynomial regression by adding new features x^2, x^3, x^4 to the inputs. Repeat (b) and (c). Compare the training error and test error. Compared with the previous results, which order is the best for fitting the data?

3 Regularization and Cross-Validation [25 points]

(a) [10 points] Using the training data to implement ℓ_2-regularized for the 4th-order polynomial regression (page 12 of Lecture 4, note that we do not penalize the bias term w_0), vary the regularization parameter $\lambda \in \{0.01, 0.05, 0.1, 0.5, 1, 100, 10^5\}$. Plot the training and test error (averaged over all instances) using Eq. (1) as a function of λ (you should use a $\log_{10}$ scale for λ). Which λ is the best for fitting the training data? Which λ is the best for fitting the test data?

(b) [10 points] Plot the value of each weight parameter (including the bias term w_0) as a function of λ.

(c) [5 points] Write a procedure that performs five-fold cross-validation on your training data (page 7 of Lecture 4). Use it to determine the best value for λ. Show the average error on the validation set as a function of λ. Is the same as the best λ in (a)? For the best fit, plot the test data and the ℓ_2-regularized 4th-order polynomial regression line obtained.